

Geometric transformations: inversion

Andrei Bud

1 Problems

Archimedes' Lemma: Let ω_1, ω_2 be two circles internally tangent at a point P . A chord AB of ω_1 is tangent to ω_2 at a point C . Then we have that PC is the angle bisector of $\angle APB$.

Ptolemy's Theorem: Let $ABCD$ be a cyclic quadrilateral. Then we have the equality

$$AB \cdot CD + AD \cdot BC = AC \cdot BD.$$

Problem 1. Let l_1 and l_2 be two perpendicular lines intersecting at O and let $\omega_1, \omega_2, \omega_3, \omega_4$ be four circles such that

- ω_1 and ω_2 are tangent to l_1 in O and
- ω_3 and ω_4 are tangent to l_2 in O .

These four circles intersect at 4 points A, B, C, D other than O . Prove that A, B, C, D are concyclic.

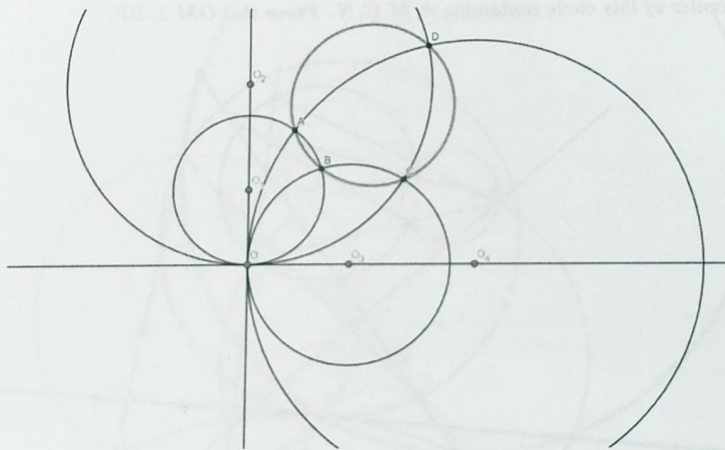


Figure 1: Configuration of the problem

Problem 2. We consider two concentric circles ω, ω' of center O and radii R and R' , with $R > R'$. Let A be a point on ω and B, C the intersection points between the circle of diameter OA and ω' . Let D be the intersection of (OA) with ω' , while E and F are the intersections of the rays $(OB), (OC)$ with ω . Prove that D, E, F are collinear.

Problem 3. Let ABC be a triangle and let D be the foot of the perpendicular from A to BC . The circle of diameter AD intersects AC and AB at points E and F . Prove that $AO \perp EF$, where O is the circumcenter of $\triangle ABC$.

Problem 4. Let $ABCD$ be a bicentric quadrilateral and let M, N, P, Q be the contact points of the incircle with its sides. Prove that $MP \perp NQ$.

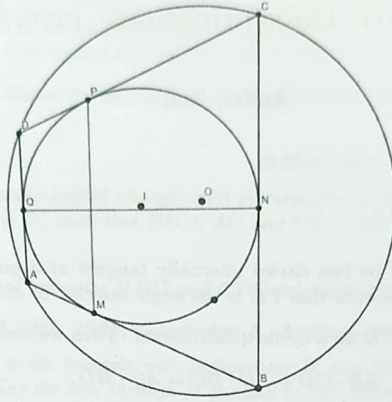


Figure 2: Configuration of the problem

Problem 5. Let $ABCDEF$ be a hexagon inscribed in a circle ω and assume that lines AB, CF and DE intersect in a point M . The circumcircle ω' of $\triangle BMD$ intersects BF in P , and we take $MP \cap AE = \{N\}$.

1. Prove that A, M, C, N are concyclic.
2. Let O be the center of this circle containing A, M, C, N . Prove that $OM \perp BF$.

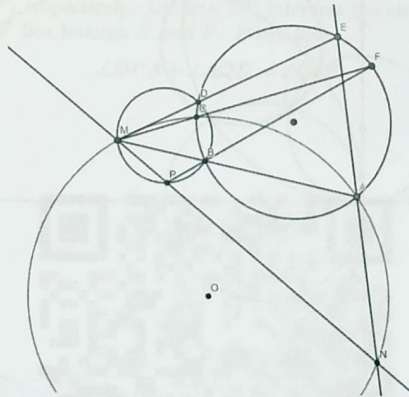


Figure 3: Configuration of the problem

Problem 6. Let ABC be a triangle, ω its circumcircle and D a point on the side BC . We consider the circles ω_1, ω_2 tangent to $\omega, (AD), (BD)$ and to $\omega, (AD), (CD)$, respectively.

Prove that ω_1 and ω_2 are tangent if and only if $\angle BAD \equiv \angle CAD$.

Problem 7. Let ABC be a triangle and M, N, P the contact points of the incircle with sides BC, AC and AB respectively. Prove that the orthocenter of $\triangle MNP$, the incircle of $\triangle ABC$ and the circumcircle of $\triangle ABC$ are collinear.

2 Inversion problems at IMO

Several geometry problems at the IMO admit solutions via inversion. In a substantive number of cases these problems are Geometry Shortlist G1-G3, and hence solving them can guarantee a bronze medal, or at least a honorable mention. Out of this list of problems we recall

Problem 9. Let $ABCD$ be a trapezoid with parallel sides $AB > CD$. Points K and L lie on the line segments AB and CD , respectively, so that $AK/KB = DL/LC$. Suppose that there are points P and Q on the line segment KL satisfying

$$\angle APB = \angle BCD \quad \text{and} \quad \angle CQD = \angle ABC.$$

Prove that the points P , Q , B and C are concyclic.

IMO SHL 2006, Geometry 2, Author: Vyacheslav Yasinskiy, Ukraine:

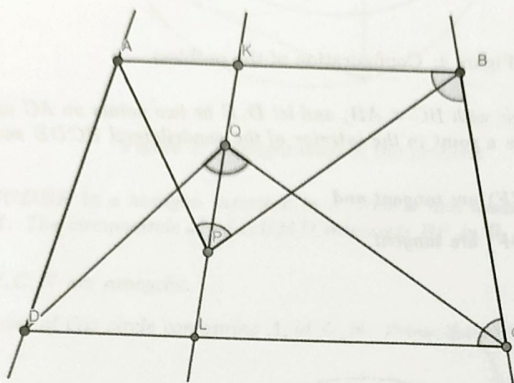


Figure 6: Configuration of the problem

Problem 10. Let ABC be an acute-angled triangle with $AB \neq AC$. The circle with diameter BC intersects the sides AB and AC at M and N respectively. Denote by O the midpoint of the side BC . The bisectors of the angles $\angle BAC$ and $\angle MON$ intersect at R . Prove that the circumcircles of the triangles BMR and CNR have a common point lying on the side BC .

IMO 2004, Problem 1

Problem 11. Let ABC be an acute triangle with D, E, F the feet of the altitudes lying on BC, CA, AB respectively. One of the intersection points of the line EF and the circumcircle is P . The lines BP and DF meet at point Q . Prove that $AP = AQ$.

IMO Shortlist 2010 - Geometry 1, Author: Christopher Bradley, United Kingdom

Problem 12. Let P be a point interior to triangle ABC (with $CA \neq CB$). The lines AP , BP and CP meet its circumcircle Γ again at K , L , and M , respectively. The tangent line at C to Γ meets the line AB at S . Show that $SC = SP$ implies $MK = ML$.

IMO 2010, Problem 4 - Author: Marcin Kuczma, Poland

Problem 13. Let $ABCD$ be a convex quadrilateral and let P and Q be points in $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point E on the line segment PQ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic.

IMO SHL 2008, Geometry 3, Proposed by John Cuya, Peru

Problem 14. Given trapezoid $ABCD$ with parallel sides AB and CD , assume that there exist points E on line BC outside segment BC , and F inside segment AD such that $\angle DAE = \angle CBF$. Denote by I the point of intersection of CD and EF , and by J the point of intersection of AB and EF . Let K be the midpoint of segment EF , assume it does not lie on line AB . Prove that I belongs to the circumcircle of ABK if and only if K belongs to the circumcircle of CDJ .

IMO Shortlist 2008, Geometry 2, Author: Charles Leytem, Luxembourg

The list of problems admitting proofs via inversion goes on and on. Out of this list we recall the problems:

- IMO Shortlist 2023, G3
- IMO Shortlist 2022, G2
- IMO Shortlist 2020, G3
- IMO Shortlist 2018, G2
- IMO Shortlist 2017, G3
- IMO Shortlist 2013, G2
- IMO 2013, Problem 4.

3 $\sqrt{bc}$ -inversion

Up to this point, we noticed that inversion changes the problem configuration in such a way that the new statement might be completely unrecognizable. There exists, however, one way to apply inversion such that the inverted problem maintains some semblance to the initial statement:

Let $\triangle ABC$ be a triangle. We can apply an inversion of center A , and radius $\sqrt{bc}$ followed by a symmetry with respect to the angle bisector AI . This transformation interchanges points B and C and points I and I_A , and we will see how it can be used in many geometry problems. We will focus especially on proving some basic properties of the mixtilinear circles.

Problem 15. Let $\triangle ABC$ be a triangle, and let ω denote its circumcircle. Let D' be the point of tangency between side BC and the A -excircircle of triangle ABC . Let X be the point of tangency between ω and the A -mixtilinear incircle (i.e. the circle tangent to ω and to segments AB and AC).

Prove that

$$\angle BAX = \angle CAD'.$$

Problem 16. Let $\triangle ABC$ be a triangle, and let ω be its circumcircle. Let D be the point of tangency between side BC and the incircle of triangle ABC . Let X be the point of tangency between ω and the A -mixtilinear circle (i.e., the circle tangent to ω and to the segments AB and AC).

Prove that

$$\angle BXA = \angle CXD.$$

Problem 17. Let $\triangle ABC$ be a triangle, and let ω be its circumcircle. Let D be the point of tangency between side BC and the incircle of triangle ABC . Let X be the point of tangency between ω and the A -mixtilinear circle (i.e., the circle tangent to ω and to the segments AB and AC). Let E be a point on the segment BC and $F = AE \cap \omega$. Prove that $XDEF$ is cyclic.

Problem 18. Let ABC be a triangle with circumcircle ω , incenter I , and A -excenter I_A . Let the incircle and the A -excircle touch BC at points D and E , respectively. Let M be the midpoint of arc BC (not containing A) of ω .

Consider the circle that is tangent to BC at D and tangent to arc BAC of ω at a point T . Let TI intersect ω again at S .

Prove that lines SI_A and ME intersect on ω .

ELMO 2010, Problem 6

Problem 19. Let $\triangle ABC$ be an acute triangle with $AB < AC$, and let H and O denote its orthocenter and circumcenter, respectively. Let Γ be the circumcircle of triangle $\triangle BOC$. The line AO intersects Γ again at A' , and the circle centered at A with radius AO intersects Γ at a point F .

Prove that Γ , the circle with diameter AA' , and the circle passing through A , F , and H , all pass through a common point.

RMM 2025, Author: Radu Andrei Iecoiu

Problem 20. Let ABC be an acute-angled triangle with circumcircle ω and circumcenter O . Points D and E lie on ω , with $D \neq B$ and $E \neq C$, such that $BD \perp AC$ and $CE \perp AB$. Let CO intersect AB at X , and let BO intersect AC at Y .

Prove that the circumcircles of triangles BXD and CYE intersect at a point that lies on line AO .

IMO SHL 2023, Geometry 5, Author: Ivan Chan Kai Chin, Malaysia

Problem 21. Let ABC be an acute triangle with orthocenter H and circumcircle Γ . Let D be the point diametrically opposite A on Γ . Let the line through H parallel to BC intersect AB and AC at points X and Y , respectively. Let AD intersect the circumcircle of triangle $DX Y$ again at point S . Let the tangent to Γ at point A intersect line XY at point T .

Prove that lines DT and HS intersect at a point on Γ .

RMM Shortlist 2024, Geometry 2

Problem 22. In triangle ABC and let D be a point on BC . The circumcircle of triangle ABD intersects segment AC again at F , and the circumcircle of triangle ACD intersects segment AB again at E . Prove that as D varies along BC , the circumcircle of triangle AEF always passes through a fixed point (distinct from A), and that this point lies on the median from A to BC .

Problem 23. Let I be the incenter of an acute-angled triangle ABC , and let the incircle touch sides BC , CA , and AB at points D , E , and F , respectively. Let line EF intersect the circumcircle of triangle ABC again at points P and Q , such that F lies between E and P . Prove that

$$\angle DPA + \angle AQD = \angle QIP.$$

IMO SHL 2019 Geometry 6



Figure 7: Solutions to the problems